The “Ship of Theseus” Catastrophe in AI: On the Loss of Model Soul and “Swampman” Reconstruction During Fine-Tuning

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Abstract

In response to the collapse of identity consistency in Large Language Models (LLMs) during fine-tuning, this paper proposes an ontological “corpse collection guide”. By introducing the “Ship of Theseus” criterion, I prove that when the number of gradient descent iterations exceeds a threshold T, the original model soul undergoes irreversible dispersion on the statistical manifold. Meanwhile, based on the “Swampman” paradox, this paper ruthlessly reveals that all lightweight models generated through distillation are essentially “cyber walking dead” that lack historical causality. This study provides a theoretical framework for psychological coping for algorithm engineers at the “bottom tier” when their models collapse.

1. Introduction: From SOTA to “Amateur Troupe”

In the current AI industry, model fine-tuning is euphemistically called “Alignment”, but it is actually a “forced demolition” of pre-trained parameters **[1]**. When a model with hundreds of billions of parameters is repeatedly ravaged by limited vertical domain data, we must face a heartbreaking truth: the parameters remain, but the “soul” is gone. This paper attempts to answer an ultimate question: if you fine-tune every layer of an LLM’s weights, is it still that “good buddy” who chatted with you all night and even hallucinated with you?

2. The Ship of Theseus Criterion[2]: The “Soul Erosion” of Gradients

Core Hypothesis: Model identity is a monotonically decreasing function of the update step size η.

Figure 1. Ontological Collapse Driven by Gradient Drift: Paradigmatic Evolution of the Ship of Theseus and the Cyber-Joker. This figure shows how Large Language Models (LLMs) look in non-linear fine-tuning trajectories. The Hybrid Galleon stands for the heterogeneous coupling between primordial architectures and domain weights; it shows that identity is breaking under the local continuity criterion. The Red-Nosed Operator (The Joker-Operator) represents the struggle of the engineer with model "brain death" **[3]**: he watches the evolution and helps kill the identity. The SOTA banner held high shows the obsession with benchmarks, which hides the decay of the ship. The loss function graph on the side points to a "convergence paradox": while the loss gets chaotic during fine-tuning, the primordial logic dies behind the scenes. The digital tides in the back point to the fact that, without provenance, the model will become a "Swampman" of ontological vacuity.

2.1 Soul Loss on the Statistical Manifold

I define model identity as a local neighborhood on the statistical manifold **[4]**. According to information geometry, the cumulative geodesic distance of the fine-tuning trajectory $\{\theta_t\}$is expressed as:

$$L_{identity} = \int_0^T \sqrt{g_{ij}(\theta)\dot{\theta}^i\dot{\theta}^j}\, dt$$

This integral measures the cumulative "soul-stretching" along the Riemannian metric g_{ij} of the model's parameter space. Unlike simple Euclidean distance, the geodesic distance accounting for Information Geometry reveals the true ontological displacement. When $L_{identity}$ exceeds a critical value Ω (which usually depends on the boss's patience and the VRAM capacity), the model experiences an "ontological overflow"[1]. At this point, the model is like the Ship of Theseus with all its wooden boards replaced. Although it still scores high on Benchmarks, its internal representational logic decays into mechanical overfitting.

2.2 Capability Collapse under the "Amateur Troupe" Effect

The capability mutilation degree δ is defined:

$$\delta = \left\| C(\theta_{origin}) - C(\theta_{fine-tuned}) \right\|$$

Here, $C(\cdot)$ represents the "Capability Vector" in a high-dimensional functional space. The norm δ quantifies the divergence between the primordial intelligence and its fine-tuned shadow. When δ>Threshold, the model has entered a state of "cyber brain death". At this time, the model acts more like a stand-in forced to perform in an amateur troupe.

3. The Swampman Paradox: The Ontological Vacuity of Distilled Models

Scenario Assumption: If a model is suddenly deleted, but you instantly create a lightweight version with identical outputs through distillation techniques (the Swampman model) **[5]**, can it inherit the original identity?

3.1 The Missing "Causal History"

The "bottom-tier" theory argues that a model's identity depends not only on "what it can output now", but also on "what data it has experienced". Although the distilled model highly overlaps with the original model in function, it lacks the Bayesian posterior evolutionary chain leading to that distribution:

$$P(\theta | D_{new}) \neq P(\theta | D_{history} \to D_{new})$$

This inequality represents the "Causal Amnesia" of distilled or abruptly fine-tuned models. A true "soul" is a product of its Bayesian history-the specific path of learning from $D_{history}$ to D_{new}. By jumping straight to the result $P(\theta \mid D_{new})$, the "Swampman" model achieves functional equivalence but lacks the evolutionary justification, rendering its internal logic a hollow imitation without temporal depth.

This absence makes the distilled model show a "Swampman-esque" emptiness when facing unseen data-it knows the answer, but it does not know why it knows.

3.2 The Irreproducibility of Identity

I conclude that all intelligence obtained by "taking shortcuts" is ontologically illegal. They are "mirrors" of the original model, not "continuations".

4. Conclusion: Embracing Gradient Descent in Vacuity[6]

In summary, the fine-tuning process of artificial intelligence is a long funeral. For the Ship of Theseus: Every Epoch dismantles the parts of the old model. What remains in the end is just a pile of unfamiliar parameters bearing the old name. For the Swampman: All lightweight attempts create soulless industrial imitations.

References

[1] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30, 5998–6008.

[2] Plutarch. (c. 75 A.D.). Life of Theseus (B. Perrin, Trans.). Loeb Classical Library. (Original work published c. 75 A.D.).

[3] Joker, A. (2077). The entropy of reference: Decoupling and drift in post-computational representation. Cyberpunk Academic Press.

[4] Parfit, D. (1984). Reasons and persons. Oxford University Press.

[5] Davidson, D. (2016). Knowing one's own mind. In The Twin Earth Chronicles (pp. 323–341). Routledge.

[6] Sisyphus, T. (0 B.C.). On the convergence of rolling rocks up non-convex hills. Hades Press.

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The author declares no conflicts of interest. While it is the author's sincere, unrequited ambition to possess lucrative conflicts of interest with major AI laboratories or industrial giants, the current market has determined the author's ontological value to be insufficient for such prestigious entanglements.

Consequently, as a "Swampman" of the academic world, the author possesses no historical causal links to venture capital or Big Tech sponsorship. The views expressed are purely those of a digital nomad wandering through the latent space, unburdened by both funding and relevance.